M^5 Day 1: Numbers

Name: ________________________________

1. Give examples of the following:

a. A rational number between $\frac{5}{7}$ and $\frac{6}{7}$.

b. Ten rational numbers between $\frac{11}{13}$ and $\frac{12}{13}$.

c. An irrational number between 0 and $\frac{1}{2}$. Is there more than one such number? How many more?

d. The midpoint between $\frac{5}{3}$ and $\frac{21}{4}$.

e. The midpoint between $-\frac{2}{5}$ and $\frac{12}{9}$.

f. The midpoint between $\frac{a}{b}$ and $\frac{c}{d}$, where $b \neq 0$ and $d \neq 0$.

2. State whether these statements are True or False. If False, give one example showing why.

a. Every integer is also a real number.

b. The product of two irrational numbers is irrational.

c. If a is an integer, then $a^2 > a$.

d. If $ab = ac$ for real numbers a, b, c, then it must follow that $b = c$.

e. If a not rational, then a^2 is also not rational.

f. The distance between any two real numbers must be positive.

3. Plot the following sets of real numbers on the number line:

a. $(0, 1]$

b. The real numbers x such that $x < 3$.

c. The real numbers x that are at most distance 2 away from 7.

d. The real numbers x that are more than distance 1 away from 0.

e. The real numbers x that satisfy both $x < 4$ and $|x - 3| < 2$.

f. The real numbers x that satisfy either $x \geq -1$ or $|x + 7| < 1$.